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Assessment-2

SECTION-A

Section A: Scenario-Based Concept Application

1. Explain how Deep Packet Inspection (DPI) at the Network and Transport layers

of the OSI model allows a startup to isolate specific traffic bottlenecks causing

slow internet speeds despite high available bandwidth.

Ans: A startup has high bandwidth available, but the internet is still slow. In this case, **Deep Packet Inspection (DPI)** examines packet contents at the **Network Layer (IP addresses)** and **Transport Layer (TCP/UDP ports)**.

- DPI identifies which applications or users are consuming excessive bandwidth.
- It can detect heavy traffic such as video streaming, large downloads, or cloud backups.
- By analyzing packet details, the network administrator can locate the exact source of congestion.
- This helps isolate bottlenecks and apply traffic control or prioritization policies.

Conclusion: DPI helps find the specific traffic causing slow performance even when total bandwidth is sufficient.

2. Explain the distinct roles of DHCP and DNS in a scenario where startup devices

are successfully assigned IP addresses but remain unable to resolve external

web addresses for internet access.

Ans: Suppose startup devices receive IP addresses successfully but cannot open websites.

DHCP (Dynamic Host Configuration Protocol)

- Automatically assigns IP addresses to devices.
- Provides network settings such as subnet mask and gateway.
- Since devices have valid IP addresses, DHCP is working correctly.

DNS (Domain Name System)

- Converts website names (e.g., google.com) into IP addresses.
- If DNS fails, users cannot access websites using names even though the internet connection exists.

Conclusion: DHCP provides the device's address, while DNS translates website names. If IP assignment works but websites cannot be reached, the DNS service is likely the problem.

3. Explain the necessity of Inter-VLAN Routing and the specific role of static

routes when a startup requires communication between two isolated

departments on the same physical switch.

Ans: A startup has two departments (Management and Staff) placed in different VLANs on the same switch.

Why Inter-VLAN Routing is Needed

- VLANs create separate logical networks.
- Devices in one VLAN cannot communicate directly with devices in another VLAN.
- A router or Layer-3 switch performs **Inter-VLAN Routing** to allow communication.

Role of Static Routes

- Static routes tell the router where packets for a particular network should be sent.
- They provide a fixed path between VLAN networks.
- Useful in small startup networks because they are simple and secure.

Conclusion: Inter-VLAN Routing enables communication between VLANs, while static routes define the path packets should follow.

4. Explain the performance differences between ICMP and HTTP/HTTPS protocols

that would cause a website to experience high latency while a ping test

returns a fast, successful response.

Ans: A startup notices that ping responses are quick, but websites load slowly.

ICMP (Ping)

- Uses small packets.
- Checks only basic network connectivity.
- Does not establish sessions or transfer web content.

HTTP/HTTPS

- Used for web browsing.
- Requires TCP connection setup.
- HTTPS also performs SSL/TLS encryption and certificate verification.
- Transfers web pages, images, videos, and scripts.

Because HTTP/HTTPS involves more processing and data transfer, latency may be high even when ICMP responses are fast.

Conclusion: Fast ping only shows good connectivity; slow website loading can result from web server delays, encryption overhead, or heavy webpage content.

5. Explain the conceptual logic of Firewall Access Control Lists (ACLs) and the

strategic placement of a firewall to restrict unauthorized traffic between a

startup's management and general staff departments.

Ans: A startup wants to prevent general staff from accessing the Management department network.

Firewall ACL (Access Control List)

- ACLs contain rules that allow or deny specific traffic.
- Rules are based on IP addresses, ports, and protocols.
- Example:
    - Allow Management → Staff traffic.
    - Deny Staff → Management traffic.

Strategic Firewall Placement

- The firewall should be placed between the Management VLAN and Staff VLAN.
- All traffic between departments passes through the firewall.
- The firewall checks ACL rules before forwarding packets.

Conclusion: ACLs define what traffic is permitted or blocked, while placing the firewall between departments ensures unauthorized access is prevented.

SECTION-B

Task 1: Design a VLSM subnet scheme for 3 departments requiring 15 hosts each using $192.168.10.0/24$.

Ans: **Step 1: Find Suitable Subnet Size**

For 15 hosts:

$$2^n - 2 \geq 15$$
$$2^5 - 2 = 30$$

So, **5 host bits** are needed.

Prefix:

$$32 - 5 = 27$$

Therefore, use **/27 subnet mask**.

Subnet Mask:

$$255.255.255.224$$

Department	Network Address	Host Range	Broadcast
VLAN 10	192.168.10.0/27	192.168.10.1 – 192.168.10.30	192.168.10.31
VLAN 20	192.168.10.32/27	192.168.10.33 – 192.168.10.62	192.168.10.63
VLAN 30	192.168.10.64/27	192.168.10.65 – 192.168.10.94	192.168.10.95

Final VLSM Scheme

- Department 1 → **192.168.10.0/27**
- Department 2 → **192.168.10.32/27**
- Department 3 → **192.168.10.64/27**

Each subnet supports **30 hosts**, which is enough for 15 hosts.

Task 2: Configure VLANs 10, 20, and 30 on a Cisco switch and assign physical interfaces to each.

Ans: **Step 1: Create VLANs**

```
Switch> enable
Switch# configure terminal

Switch(config)# vlan 10
Switch(config-vlan)# name Management

Switch(config-vlan)# exit
Switch(config)# vlan 20
Switch(config-vlan)# name Staff

Switch(config-vlan)# exit
Switch(config)# vlan 30
Switch(config-vlan)# name Sales

Switch(config-vlan)# exit
```

Step 2: Assign Ports to VLAN 10

Switch(config)# interface range fa0/1 - 4
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 10
Switch(config-if-range)# exit

Step 3: Assign Ports to VLAN 20

Switch(config)# interface range fa0/5 - 8
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 20
Switch(config-if-range)# exit

Step 4: Assign Ports to VLAN 30

Switch(config)# interface range fa0/9 - 12
Switch(config-if-range)# switchport mode access
Switch(config-if-range)# switchport access vlan 30
Switch(config-if-range)# exit

Step 5: Verify Configuration

Switch# show vlan brief

Expected output:

VLAN	Name	Ports
10	Management	Fa0/1–Fa0/4
20	Staff	Fa0/5–Fa0/8
30	Sales	Fa0/9–Fa0/12

Task 3: Set up a DHCP pool on a Router/L3 switch to provide dynamic addressing for

the VLAN 10 subnet.

Ans: **Step 1: Exclude Gateway Address**

Router(config)# ip dhcp excluded-address 192.168.10.1

This prevents the router from assigning the gateway IP to clients.

Step 2: Create DHCP Pool

Router(config)# ip dhcp pool VLAN10

Router(dhcp-config)# network 192.168.10.0 255.255.255.224

Router(dhcp-config)# default-router 192.168.10.1

Router(dhcp-config)# dns-server 8.8.8.8

Router(dhcp-config)# exit

Step 3: Verify DHCP Configuration

Router# show ip dhcp pool

Router# show ip dhcp binding

Working

1. PC sends DHCP Discover.
2. Router replies with DHCP Offer.
3. PC requests the offered address.
4. Router acknowledges and assigns IP.

This is called the **DORA Process**:

- Discover
- Offer
- Request
- Acknowledge

Task 4: Capture a network trace in Wireshark and identify the IP addresses in the DNS query and HTTP GET headers.

Ans:

Step 1: Start Capture

1. Open Wireshark.
2. Select the active network interface.
3. Click **Start Capture**.
4. Open a website (e.g., www.google.com).

Step 2: Find DNS Query

Filter:

dns

Example packet:

Source IP: 192.168.10.5
Destination IP: 8.8.8.8

Standard query A www.google.com

IP Addresses Identified

Field	Address
Client IP	192.168.10.5
DNS Server IP	8.8.8.8

Step 3: Find HTTP GET Request

Filter:

http.request.method == "GET"

Example packet:

GET / HTTP/1.1
Host: www.example.com

Packet details:

Source IP: 192.168.10.5
Destination IP: 93.184.216.34

IP Addresses Identified

Field	Address
Client IP	192.168.10.5
Web Server IP	93.184.216.34

HTTP GET Header Example

GET / HTTP/1.1
Host: www.example.com
User-Agent: Chrome
Accept: text/html

Important Fields

- **GET /** → Requests the webpage.
- **Host** → Website being accessed.
- **Source IP** → Client computer.
- **Destination IP** → Web server.

Viva / Exam Explanation

DNS Query

- DNS translates domain names into IP addresses.
- Client sends query to DNS server.

- Wireshark shows Source IP (client) and Destination IP (DNS server).

HTTP GET

- Browser sends GET request to obtain a webpage.
- Wireshark shows Source IP (client) and Destination IP (web server).
- The Host field indicates which website is being requested.

SECTION- C: Integrated Case Study

Scenario: TechLaunch, a startup with 40 employees, is experiencing "flapping" network connectivity. Employees in the Marketing wing report that their internet drops for 30 seconds every few minutes. The Engineering wing, located on a different VLAN, remains perfectly stable. A ping test from a Marketing PC to the Core Switch shows a consistent 25% packet loss and fluctuating latency.

Questions:

1. Based on the symptoms, identify the two most likely OSI layers where this fault is occurring.

Ans: The fault is most likely at the **Physical Layer (Layer 1)** or **Data Link Layer (Layer 2)** because packet loss and intermittent connectivity are common symptoms of cable faults, switch port errors, or switching loops.

2. Outline a 4-step "Bottom-Up" troubleshooting sequence to isolate this specific issue.

Ans: **Step 1: Check Physical Connections**

- Verify cables
- Check link LEDs
- Replace suspected cable
- Check switch port status

Step 2: Check Switch Port Statistics

- Look for CRC errors
- Check packet drops
- Check interface utilization

Step 3: Verify VLAN Configuration

- Ensure Marketing devices are in the correct VLAN
- Verify trunk links

- Check VLAN membership

Step 4: Test Connectivity

- Ping Core Switch
- Ping Gateway
- Capture packets using Wireshark
- Check for excessive broadcasts

Bottom-Up Sequence:

1. Physical cables and ports
2. Interface statistics
3. VLAN configuration
4. Connectivity testing and packet analysis

3. List three CLI commands you would run on the switch to check for cable faults or broadcast storms.

Ans: **Check Interface Errors**

show interfaces

Shows:

- CRC errors
- Input errors
- Packet drops

Check VLAN Status

show vlan brief

Shows:

- VLAN assignments
- Active VLANs

Check STP Status

show spanning-tree

Shows:

- Blocked ports
- STP state
- Potential loops

4. If the issue is a "Switching Loop," explain which protocol should have prevented

this and why it might have failed.

Ans: **Protocol**

Spanning Tree Protocol (STP)

Purpose

STP prevents switching loops by blocking redundant paths.

Without STP:

- Frames circulate endlessly.
- Broadcast traffic multiplies rapidly.
- Broadcast storm occurs.
- Network becomes unstable.

Why STP Might Fail

- STP disabled
- Incorrect STP configuration
- Misconfigured trunk links
- Recently added switch not running STP correctly

Answer:
A switching loop should be prevented by **STP (Spanning Tree Protocol)**. The issue may occur if STP is disabled, improperly configured, or fails to block redundant paths.

5. Propose a long-term configuration change to ensure an error in one

department doesn't impact the rest of the office

Ans: **Improve Network Segmentation**

Create separate VLANs for departments:

- VLAN 10 → Marketing
- VLAN 20 → Engineering
- VLAN 30 → Management

Use:

- Inter-VLAN routing through Layer-3 switch
- ACLs to control traffic
- STP on all switches

Benefit

If a broadcast storm or loop occurs in Marketing:

- It remains confined to VLAN 10.
- Engineering VLAN remains unaffected.
- Overall network stability improves.

Answer:
Implement proper VLAN segmentation with STP enabled and controlled Inter-VLAN Routing. This isolates faults within a department so that problems in one VLAN do not affect the rest of the office.